



Ford is developing a smart headlight system that can highlight pedestrians, cyclists and animals in an attempt to make driving at night safer (Image courtesy: <http://innovationworkshop.fordpresskits.com>)

Making cities safe, eco-friendly

According to a recent Navigant Research report, 40 per cent of a city's power usage can be attributed to street lighting. While this is one of the biggest reasons compelling corporations to retrofit streetlights with LEDs, the ability to make these lights smart is also working in their favour. Smart street-lighting systems offer dashboards that provide a bird's-eye view of the whole city's lighting system as well as the ability to control it. These systems also offer various ways to save power by, say, switching off and on at programmed times.

One step ahead is Philips Lumi-Motion lamp-post that combines a streetlight with a motion sensor to automatically switch on when someone approaches. Such smart systems can be combined with navigational headsets for visually-impaired people to help them understand their environment in various ways, say, by reconstructing their surroundings in sound or haptic formats. This makes smart streetlights cost-effective, controllable and socially-friendly, and saves energy, too.

This means that the use of LED lights for street lighting does not end with retrofitting. Countries have to research and completely understand their potential to ensure that these benefit the society, too.

Denmark, for example, has undertaken a study at Danish Outdoor Lighting Lab, or DOLL, an industrial

park outside Copenhagen, Denmark, with 9km of road and bike paths fitted with smart lighting systems from 18 companies. They are investigating options like controlling the system using iPads, making lights brighter when a vehicle or pedestrian approaches, managing parking and transmitting information about traffic, weather or air quality.

Adding a dimension to LEDs

All this smart lighting requires a lot of LEDs. The cost at which lights sell today makes it seem impossible to fit the whole world with it. So while probing a bit into the core tech, we stumbled upon a four-year-old French company called Aleidia that develops low-cost LEDs with a unique three-dimensional (3D) architecture. Their special WireLED microwire technology enables the manufacture of LEDs at 25 per cent the cost of current technologies by growing high-density, coaxial gallium-nitride (GaN) microwires directly onto large-diameter silicon wafers. This can be done using existing complementary metal-oxide semiconductor (CMOS) foundries.

Aleidia's technology uses economical silicon wafers with large diameters of 20cm or more and millions of vertical GaN microwires, or microrods with a diameter of less than one micron are grown on each wafer. Each of these microwires is an LED capable of emitting light from all sides. Use of large substrates significantly brings

down the cost of each 3D LED, compared to today's planar ones.

Moreover, this process also involves lesser time. The advantage of being able to manufacture and package the LEDs in existing CMOS facilities does away with large capital investments, making the new technology all the more attractive.

Cost aside, WireLED chips are also believed to be technically superior because these are designed to contain microwires that emit light over a broad range of different wavelengths, making it possible to produce green or red LEDs using the same material as industry-standard blue GaN LEDs. By combining different microwires on the same chip, white LEDs can also be manufactured without using phosphors.

Lighting without electricity

With the amount of research and development happening across the world, it is clear that smartlights are the future, be it at home, in industries or public places.

Much into the future, lighting might get even smarter—with priorities moving away from smartness and control to simply saving our planet. Dutch designer Daan Roosegaarde, for example, has come up with a completely off-beat approach to public lighting. Learning from bioluminescence in natural systems, he has found a way to light up a smart highway in the Netherlands using glow-in-the-dark paint as road markings. The special paint absorbs sunlight during the day to glow in the dark at night. In collaboration with State University of New York, USA, he has also developed a prototype plant containing luciferin, the compound that produces light in creatures like fireflies and jellyfish.

With his team, he is now working on developing bioluminescent street trees that will produce a soft-glow in the dark. When streetlights are replaced with these trees, these will not only work without electricity—a fact too good to be true—but also make the highway look like a scene out of a fairy tale! ●